PAPER CODE: BCE-P624
SEMESTER EXAMINATION (MAY-2026)
CLASS: B.TECH. SEMESTER: VI
SUBJECT: CSE
PAPER TITLE: MACHINE LEARNING - 2

Time: 3 Hours **Max. Marks: 70**

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any ***five*** questions in about 150 words each. Each question carries six marks. **(5 x 6 = 30 Marks)**

		CO	BL
Q-1	What is the difference between **regression** and **classification** in Machine Learning? Give examples.	CO2	B2
Q-2	(a)Define the Sigmoid Function. • Calculate its output for z=0, z=−2 and z=−2. (b) Given input vector is z=[2,1,0] Find 1. Compute the **softmax probabilities** for each element. 2. Verify that the sum of probabilities equals 1.	CO2	B3
Q-3	What is Clustering and how is it different from classification?	CO4	B2
Q-4	Explain how **Random Forests** reduce overfitting compared to a single decision tree.	CO2	B2
Q-5	Explain the top-down approach for constructing a decision tree.	CO3	B3
Q-6	Explain the concept of a **Decision Tree**. What are **splitting criteria** commonly used?	CO2	B2
Q-7	Differentiate between **linear SVM** and **non-linear SVM**.	CO2	B2
Q-8	Explain the steps of an **end-to-end machine learning model development pipeline**.	CO2	B2
Q-9	What is **DBSCAN**? How does it differ from K-Means and hierarchical clustering?	CO5	B2
Q-10	Write a Python program to perform classification using the K-Nearest Neighbours (KNN) algorithm. Tasks: 1. Load the Iris dataset using sklearn.datasets. 2. Split the dataset into training (70%) and testing (30%) sets. 3. Train a KNN classifier with K = 3. 4. Predict class labels for the test dataset. 5. Calculate and print the accuracy of the model. Note: - Use train_test_split from sklearn.model_selection - Use KNeighborsClassifier from sklearn.neighbors - Use accuracy_score from sklearn.metrics	CO4	B3

AB: NON-NEP

SECTION-B (LONG ANSWER TYPE QUESTIONS) **Instructions**: Answer any ***four*** questions in detail. Each question carries 10 marks. (4 x 10 = Marks)		CO	BL
Q-1	Define Linear Regression and write its hypothesis equation for one variable.	**CO1**	**B2**
Q-2	(a)Define **log loss (cross-entropy loss)** and explain why it is used instead of mean squared error. (b)Given: • Predicted probability = 0.8, actual label = 1 • Compute the log loss.	**CO5**	**B4**
Q-3	Given predicted probabilities and actual class labels: (see table below) Find: 1. Compute TPR and FPR at different thresholds. 2. Plot the ROC curve points. 3. Calculate the AUC using the trapezoidal rule.	**CO3**	**B4**
Q-4	Given the following dataset: (see table below) A new point **P(3, 2)** is to be classified. **Tasks:** 1. Compute Euclidean distance between P and all data points. 2. Find the 3 nearest neighbors (K = 3). 3. Determine the class of P using K-NN. 4. Repeat classification using **weighted KNN** (weight = 1/distance).	**CO3**	**B4**
Q-5	Given the following dataset:	**CO4**	**B4**

Q-3 table:

Sample	Probability	Actual
A	0.85	1
B	0.75	0
C	0.65	1
D	0.55	0
E	0.45	1

Q-4 table:

Point	x1	x2	Class
A	1	2	C1
B	2	3	C1
C	3	3	C2
D	6	5	C2
E	7	7	C2

Sample	Outlook	Temperature	PlayTennis
1	Sunny	Hot	No
2	Sunny	Hot	No
3	Overcast	Hot	Yes
4	Rain	Mild	Yes
5	Rain	Cool	Yes
6	Rain	Cool	No
7	Overcast	Cool	Yes
8	Sunny	Mild	No
9	Sunny	Cool	Yes
10	Rain	Mild	Yes

Tasks:

1. Calculate **Entropy** of the target variable PlayTennis.
2. Compute **Information Gain** for the attribute Outlook.
3. Determine the **root node** of the decision tree.
4. Show the first level of the decision tree after splitting by the best attribute.

Q-6 Discuss how to **evaluate and deploy a machine learning model** in a real-world scenario (e.g., web app or automated prediction system). **CO4** **B3**

Q-7 Write a Python program to build and evaluate regression models. **CO5** **B4**

Tasks:

1. Load the California Housing dataset using sklearn.datasets.
2. Perform preprocessing:
   - Handle missing values (if any)
   - Scale features using StandardScaler
3. Split the dataset into training and testing sets.
4. Train the following models:
   a) Linear Regression
   b) Random Forest Regressor
5. Predict outputs on test data.
6. Evaluate both models using:
   - RMSE (Root Mean Squared Error)
   - R^2 Score
7. Compare both models and states which performs better.

Q-8 Write a Python program to implement Linear Regression using the Least Squares Method and visualize the result. **CO6** **B4**

Tasks:

1. Create a simple dataset (or use a sample dataset).
2. Fit a linear regression model using least squares method.
3. Predict output values.
4. Plot:
   - Original data points (scatter plot)
   - Best-fit regression line
5. Label axes and add titles to the graph.

Use:
- numpy for calculations
- matplotlib for visualization

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